Supplementary Materials for
"Intrinsic Disorder Predisposition of the Lujo virus glycoprotein: A geometric algebra framework for viral protein characterization"

Supplementary Tables

Table S1. Descriptive statistics for all 18 protein groups analyzed in this study.

Internal cohesion (mean intra-group wedge product ± SD), mean standard deviation across the 16 PIM components, adaptive similarity threshold (5th percentile of intra-group wedge products; for $n = 1$ groups, threshold = 0.99), and number of sequences processed per group.

Table 1: Descriptive statistics for all 18 protein groups.

Group	Metric Sim.	Charge Angle (°)	Clifford Dist.	Total Proc.
CPP	0.95	29.44	0.20	100
NON_CPP	0.89	27.51	0.05	71
UNFOLDED	0.96	11.98	0.07	106
JUNV	0.97	44.77	0.04	3
LASV	0.96	4.25	0.05	4
PARTIALLY_FOLDED	0.99	21.25	0.03	153
LCMV	0.97	2.50	0.05	3
SIGNALS	0.98	5.20	0.06	47
VIRUS_UNREVIEWED	0.99	6.52	0.02	1,087,163
UNREVIEWED_ALL	0.99	7.09	0.04	149,234,636
VIRUS_REVIEWED	0.99	3.39	0.03	17,517
UNREVIEWED_HUMAN	1.00	8.71	0.03	127,090
REVIEWED_HUMAN	0.99	5.86	0.03	20,416
MEMBRANE	1.00	7.81	0.02	165
REVIEWED_ALL	1.00	4.21	0.02	575,503
DISEASE	1.00	8.25	0.03	3,942
MACV	0.99	5.72	0.02	3

Table S2. Ranked comparison of all 17 protein groups against the LUJV glycoprotein centroid.

Ordered from highest to lowest wedge similarity. Values represent wedge similarity (mean ± SD from bootstrap resampling with 100 iterations), wedge orientation (+1 = same orientation, −1 = opposite orientation), and hydrophobic rotation angle (degrees).

Table 2: Ranked comparison against LUJV glycoprotein centroid.

Compared Group	**Wedge Sim.**	**Orientation**	**Hydro Angle (°)**
CPP	0.12	-1	26.45
NON_CPP	0.07	-1	28.51
UNFOLDED	0.06	-1	12.92
JUNV	0.04	1	2.68
LASV	0.09	-1	15.20
LCMV	0.06	-1	13.79
PARTIALLY_FOLDED	0.04	1	6.70
UNREVIEWED_ALL	0.02	1	1.98
SIGNALS	0.05	1	8.99
VIRUS_REVIEWED	0.03	1	5.91
VIRUS_UNREVIEWED	0.02	1	5.80
REVIEWED_ALL	0.03	1	2.01
DISEASE	0.02	1	3.24
MEMBRANE	0.02	1	0.90
UNREVIEWED_HUMAN	0.02	1	3.81
REVIEWED_HUMAN	0.03	1	4.25
MACV	0.04	-1	5.62

Table S4. Top 20 individual human proteins most similar to the LUJV glycoprotein centroid.

Ranked by wedge similarity. Columns: Rank (order of similarity), Group (protein set of origin), Protein ID (UniProt identifier), Wedge Similarity (functional similarity score; 1 = identical, 0 = orthogonal), Orientation (+1 = same directional organization, −1 = opposite), and Hydro Angle (°) (rotation angle in the hydrophobic plane; smaller angles indicate better alignment of hydrophobic interaction patterns).

Table 3: Pairwise wedge similarity matrix between centroids of all 18 protein groups.

Values range from 0 to 1, where higher values indicate greater functional similarity between polarity profiles. The matrix is symmetric about the diagonal (differences ≤ 0.001), confirming computational consistency. Abbreviations: PF = PARTIALLY_FOLDED; RH = REVIEWED_HUMAN; UH = UNREVIEWED_HUMAN; VR = VIRUS_REVIEWED; VU = VIRUS_UNREVIEWED; RA = REVIEWED_ALL; UA = UNREVIEWED_ALL.

Group	LUJV	LASV	JUNV	MACV	LCMV	PF	CPP	NCPP	UNF	RH	UH	SIG	MEM	DIS	VR	VU	RA	UA
LUJV	0.00	0.09	0.04	0.03	0.06	0.04	0.20	0.14	0.06	0.02	0.02	0.05	0.01	0.02	0.03	0.03	0.02	0.03
LASV	0.09	0.00	0.11	0.05	0.03	0.05	0.19	0.08	0.03	0.07	0.08	0.06	0.10	0.08	0.07	0.06	0.11	0.11
JUNV	0.04	0.11	0.00	0.06	0.08	0.07	0.23	0.16	0.09	0.04	0.03	0.06	0.02	0.03	0.04	0.04	0.02	0.02
MACV	0.03	0.05	0.06	0.00	0.03	0.02	0.18	0.10	0.03	0.02	0.02	0.03	0.04	0.03	0.02	0.02	0.05	0.05
LCMV	0.06	0.03	0.08	0.03	0.00	0.02	0.19	0.09	0.01	0.04	0.05	0.03	0.07	0.05	0.04	0.04	0.07	0.07
PF	0.04	0.05	0.07	0.02	0.02	0.00	0.17	0.09	0.02	0.03	0.03	0.02	0.05	0.04	0.02	0.03	0.06	0.06
CPP	0.20	0.19	0.23	0.18	0.19	0.17	0.00	0.13	0.15	0.19	0.20	0.18	0.20	0.20	0.20	0.20	0.20	0.20
NCPP	0.14	0.08	0.16	0.10	0.09	0.09	0.13	0.00	0.07	0.12	0.13	0.10	0.14	0.13	0.12	0.12	0.15	0.14
UNF	0.06	0.03	0.09	0.03	0.01	0.02	0.15	0.07	0.00	0.04	0.05	0.03	0.07	0.05	0.04	0.04	0.07	0.07
RH	0.02	0.07	0.04	0.02	0.04	0.03	0.19	0.12	0.04	0.00	0.01	0.03	0.03	0.01	0.01	0.01	0.03	0.03
UH	0.02	0.08	0.03	0.02	0.05	0.03	0.20	0.13	0.05	0.01	0.00	0.03	0.02	0.00	0.01	0.01	0.03	0.03
SIG	0.05	0.06	0.06	0.03	0.03	0.02	0.18	0.10	0.03	0.03	0.03	0.00	0.06	0.04	0.03	0.03	0.06	0.05
MEM	0.01	0.10	0.02	0.04	0.07	0.05	0.20	0.14	0.07	0.03	0.02	0.06	0.00	0.02	0.03	0.03	0.01	0.02
DIS	0.02	0.08	0.03	0.03	0.05	0.04	0.20	0.13	0.05	0.01	0.00	0.04	0.02	0.00	0.01	0.01	0.03	0.03
VR	0.03	0.07	0.04	0.02	0.04	0.02	0.20	0.12	0.04	0.01	0.01	0.03	0.03	0.01	0.00	0.00	0.04	0.04
VU	0.03	0.06	0.04	0.02	0.04	0.03	0.20	0.12	0.04	0.01	0.01	0.03	0.03	0.01	0.00	0.00	0.04	0.04
RA	0.02	0.11	0.02	0.05	0.07	0.06	0.20	0.15	0.07	0.03	0.03	0.06	0.01	0.03	0.04	0.04	0.00	0.01
UA	0.03	0.11	0.02	0.05	0.07	0.06	0.20	0.14	0.07	0.03	0.03	0.05	0.02	0.03	0.04	0.04	0.01	0.00

Table 4: Top 20 individual human proteins most similar to LUJV.

Rank Hydro Angle (°)	Group	Protein ID	Wedge Sim.	Orientation
1 90.0	PARTIALLY_FOLDED	sp—PO9999—	0.551	-1
2 90.0	PARTIALLY_FOLDED	sp—PO9999—	0.526	1
3 40.08	PARTIALLY_FOLDED	sp—PO9999—	0.509	1
4 40.08	PARTIALLY_FOLDED	sp—PO9999—	0.420	-1
5 49.92	PARTIALLY_FOLDED	sp—PO9999—	0.400	1
6 40.08	PARTIALLY_FOLDED	sp—PO9999—	0.398	1
7 40.08	PARTIALLY_FOLDED	sp—PO9999—	0.396	1
8 49.92	PARTIALLY_FOLDED	sp—PO9999—	0.393	-1
9 40.08	REVIEWED_ALL	sp—P35904— (CDHR5)	0.389	1
10 40.08	CPP	—	0.381	-1
11 40.08	REVIEWED_ALL	sp—P0DJB0— (NPAS3)	0.375	1
12 49.92	CPP	—	0.374	-1
13 49.92	SIGNALS	sp—Q156A1—	0.373	-1
14 49.92	PARTIALLY_FOLDED	sp—PO9999—	0.372	-1
15 49.92	PARTIALLY_FOLDED	sp—PO9999—	0.372	-1
16 49.92	CPP	—	0.371	-1
17 49.92	CPP	—	0.371	-1
18 49.92	CPP	—	0.371	-1
19 49.92	CPP	—	0.371	-1
20 49.92	CPP	—	0.371	-1

Table S5. Polarity transition frequency matrix for the 30-residue example peptide.

Example sequence: AIMTAAINCSEETCSAIMTAAINCSEETCS.

Table 5: Polarity transition frequency matrix for the example peptide.

From \ To	**P^+**	**P^-**	**N**	**NP**
P^+	0	0	0	0
P^-	0	2	2	0
N	0	1	8	2
NP	0	0	4	10

Supplementary Methods

Detailed SGP Algorithm Description

The SGP (Specular Grassmann-PIM) algorithm processes protein sequences through the following steps:

Step 1 — Polarity classification

Each amino acid is mapped to one of four polarity/charge categories according to the PIM 3.0v scheme:

- Positively charged: $P^+ = \{H, K, R\}$
- Negatively charged: $P^- = \{D, E\}$
- Neutral polar: $N = \{C, G, N, Q, S, T, Y\}$
- Non-polar (hydrophobic): $NP = \{A, F, I, L, M, P, V, W\}$

Step 2 — Construction of the polarity string

The polarity symbol of each residue in a sequence is used as a replacement. Example sequence (30 residues):
[A, I, M, T, A, A, I, N, C, S, E, E, T, C, S, A, I, M, T, A, A, I, N, C, S, E, E, T, C, S]
Polarity string:
[NP, NP, NP, N, NP, NP, NP, N, N, N, P^+, P^+, N, N, N, NP, NP, NP, N, NP, NP, NP, N, N, N, P^+, P^+, N, N, N].

Step 3 — Counting adjacent transitions

Every consecutive pair of residues contributes to a 4×4 transition matrix (rows = "from", columns = "to"). For the example, total transitions = 29. The resulting transition frequency matrix is shown in **Table S5**.

Step 4 — Normalisation and vectorisation

Each count is divided by the total number of transitions (29), resulting in a 16-component probability vector (the PIM profile). The 16 components are obtained by unrolling the matrix row by row.

Step 5 — Grassmann wedge product (functional similarity)

The wedge product of two vectors v and w is the cosine of the angle between them:

$$v \wedge w = \frac{v \cdot w}{\|v\|\|w\|}$$

A value of 1 means identical functional profiles and a value of 0 orthogonality.

Step 6 — Rotor angles (Clifford rotor analysis)

The vectors are projected onto six biologically relevant planes (e.g., the hydrophobic plane defined by components N→N and NP→NP). The rotation angle θ required to align the two projections is computed as:

$$\theta = \arccos\left(\frac{p_1 \cdot p_2}{\|p_1\|\|p_2\|}\right) \text{ (in degrees)}$$

Step 7 — Specular reflection detection

A "reflected" version of vector v is created by swapping the roles of P^+ and P^- (indices $0 \leftrightarrow 5$, $1 \leftrightarrow 4$, $2 \leftrightarrow 6$, $3 \leftrightarrow 7$, etc.) and applying the real geometric algebra reflection formula:

$$v' = v - 2(v \cdot n)n$$

If the wedge product between this reflected vector and a target vector w exceeds 0.95, the two proteins are considered specular reflections of each other, indicating a charge-swapping functional relationship.

Step 8 — Clifford signature

For each group centroid, a compact 6-number fingerprint is calculated, comprising:

- Norm (vector magnitude)
- Auto-reflection (similarity with its own charge-swapped version)
- Hydrophobic projection (norm of components 10 and 15)
- Charge projection (norm of components 0 and 5)
- Auto-rotation (wedge product with a 4-position rolled copy)
- Total bivector (average of selected bivector components)

Step 9 — Adaptive threshold and anomaly detection

The similarity threshold for each group is calculated automatically as the 5th percentile of intra-group wedge products. A query protein is classified as an anomaly if its best match has a wedge product below the adaptive threshold of its closest group and a Mahalanobis distance-based probability of belonging lower than 0.5.

Supplementary Figure

Figure S1. Additional histograms of wedge product similarity distributions

Additional histograms of wedge product similarity distributions between LUJV and specific protein subgroups are available from the corresponding author upon request.

Supplementary Materials availability: All supplementary tables, methods, and figures are available in this document. Additional data and custom programs are available from the corresponding author upon reasonable request.